

Block-Diagonal Boundary Conditions in the Degenerate Parent-Hamiltonian Ground Space

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1 Notation

For a tensor $A = (A_i)_{i=1}^d$ with bond dimension D , write

$$\Gamma_N^A(X)(i_1, \dots, i_N) = \text{tr}(A_{i_1} \cdots A_{i_N} X) \quad (X \in M_D(\mathbb{C})). \quad (1)$$

Set

$$\mathcal{G}_N^A = \{\Gamma_N^A(X) : X \in M_D(\mathbb{C})\}, \quad V^{(N)}(A) = \Gamma_N^A(I). \quad (2)$$

For the cyclic chain, with $\text{res}_{s,L}$ denoting restriction to the length- L interval beginning at $s \in \mathbb{Z}/N\mathbb{Z}$, the local ground-space condition is

$$\mathcal{K}_{N,L}(A) = \bigcap_{s \in \mathbb{Z}/N\mathbb{Z}} \{\psi : \text{res}_{s,L}(\psi) \in \mathcal{G}_L^A\}. \quad (3)$$

2 The source assertion

Cirac et al. describe the degenerate parent-Hamiltonian ground space for a matrix product state with more than one block in canonical form in Section IV.C of [CPGSV21].¹ The source's block-injective canonical form is a block-diagonal physical-virtual correspondence. If

$$\tilde{A}^i = \bigoplus_{j=1}^g A_j^i, \quad (4)$$

then

$$\forall X = (X_j)_j \in \bigoplus_{j=1}^g M_{D_j}(\mathbb{C}), \quad \exists (c_i(X))_i, \quad X = \sum_i c_i(X) \tilde{A}^i. \quad (5)$$

¹For traceability, this note corresponds to issue #905 and issue #870. The relevant source lines are `Papers/2011.12127/TN-Review-main.tex:2112–2128`.

After blocking, the corresponding word-span statement is

$$\exists m_0 \quad \forall m \geq m_0, \quad \text{span}\{(A_1^\omega, \dots, A_g^\omega) : |\omega| = m\} = \prod_{j=1}^g M_{D_j}(\mathbb{C}). \quad (6)$$

The resulting parent-Hamiltonian theorem in [CPGSV21, Section IV.C] is

$$N \geq L_0 + 1 \implies \ker(H_N) = \text{span}\{V^{(N)}(A_j) : j = 1, \dots, g\}. \quad (7)$$

The sentence preceding the theorem fixes the boundary form used in the proof:

$$X = \bigoplus_{j=1}^g X_j, \quad Y = \bigoplus_{j=1}^g Y_j \quad (X_j, Y_j \in M_{D_j}(\mathbb{C})). \quad (8)$$

3 The block-diagonal intersection identity

The older representation paper [PGVWC07, Theorem *Degeneracy of the ground space v2*] gives the finite-chain identity used by the source argument.² It fixes the block count and local length by

$$b = g, \quad L \geq 3(b-1)(L_0+1) + 1. \quad (9)$$

Thus

$$m \geq L \quad (10)$$

is the range used in the intersection step:

$$\mathbb{C}^d \otimes \left(\bigoplus_j \mathcal{G}_m^{A_j} \right) \cap \left(\bigoplus_j \mathcal{G}_m^{A_j} \right) \otimes \mathbb{C}^d = \bigoplus_j \mathcal{G}_{m+1}^{A_j}. \quad (11)$$

For a vector ψ in the left-hand side,

$$\begin{aligned} \psi &= \sum_j \sum_{i_1, \dots, i_{m+1}} \text{tr} \left(A_{i_{m+1}}^j C_{i_1}^j A_{i_2}^j \cdots A_{i_m}^j \right) |i_1 \cdots i_{m+1}\rangle \\ &= \sum_j \sum_{i_1, \dots, i_{m+1}} \text{tr} \left(D_{i_{m+1}}^j A_{i_1}^j A_{i_2}^j \cdots A_{i_m}^j \right) |i_1 \cdots i_{m+1}\rangle. \end{aligned} \quad (12)$$

The direct-sum separation lemma and the one-block injectivity condition identify, for every j, i_1, i_{m+1} ,

$$A_{i_{m+1}}^j C_{i_1}^j = D_{i_{m+1}}^j A_{i_1}^j. \quad (13)$$

With

$$E^j = \sum_{i_1} C_{i_1}^j A_{i_1}^j, \quad (14)$$

²Source location: `Papers/quant-ph_0608197/MPSarchive.tex:1424–1456`.

and the normalization

$$\sum_{i_1} A_{i_1}^j A_{i_1}^{j\dagger} = I, \quad (15)$$

the same proof concludes

$$A_{i_{m+1}}^j C_{i_1}^j = A_{i_{m+1}}^j E^j A_{i_1}^j, \quad (16)$$

which puts the vector in $\bigoplus_j \mathcal{G}_{m+1}^{A_j}$.

4 The remaining boundary condition

The live blueprint records the analogous statement in source notation, without claiming that it is proved.³ Write

$$\tilde{A}_i = \bigoplus_{j=1}^b \mu_j A_i^j \quad (17)$$

and

$$\sum_{j=1}^b \mathcal{K}_{N,L}(A_j) = \left\{ \sum_{j=1}^b \psi_j : \psi_j \in \mathcal{K}_{N,L}(A_j) \right\}. \quad (18)$$

The missing boundary-condition statement is then

$$[(\forall j, \mu_j \neq 0) \wedge L \geq 3(b-1)(L_0+1)+1 \wedge N \geq L+L_0] \implies \mathcal{K}_{N,L}(\tilde{A}) = \sum_{j=1}^b \mathcal{K}_{N,L}(A_j). \quad (19)$$

The corresponding blueprint entry records the weight hypothesis but weaker length hypotheses:

$$\forall j, \mu_j \neq 0, \quad 1 < L, \quad N \geq L+1. \quad (20)$$

For

$$X = \bigoplus_j X_j, \quad (21)$$

³Blueprint locations: `def:parent_hamiltonian_ground_space_bnt` for the block-sum chain ground space, `lem:isup_chain_ground_space_block_le_assembled` for the forward inclusion, `lem:conditional_boundary_matrix_block_split_projection_span` for the direct-sum boundary formula, and `thm:gs_eq_bnt_span` for the final span statement in `blueprint/src/chapter/ch14_parent_hamiltonian.tex`.

the boundary-matrix formula is

$$\begin{aligned}\Gamma_N^{\tilde{A}}(X)(i_1, \dots, i_N) &= \sum_j \mu_j^N \operatorname{tr}(A_{i_1}^j \cdots A_{i_N}^j X_j) \\ &= \sum_j \mu_j^N \Gamma_N^{A_j}(X_j)(i_1, \dots, i_N).\end{aligned}\quad (22)$$

For the next display, abbreviate

$$\Gamma_j(X) = \Gamma_N^{A_j}(X), \quad V_j = V^{(N)}(A_j). \quad (23)$$

Thus the desired periodic-chain proof is the composition of the two implications

$$\begin{aligned}\psi \in \mathcal{K}_{N,L}(\tilde{A}) &\implies \exists (X_j)_{j=1}^b, \quad \psi = \sum_{j=1}^b \Gamma_j(X_j), \\ \forall 1 \leq j \leq b, \quad \Gamma_j(X_j) &\in \mathcal{K}_{N,L}(A_j),\end{aligned}\quad (24)$$

and, for every $X_1, \dots, X_b$,

$$\begin{aligned}\forall 1 \leq j \leq b, \quad \Gamma_j(X_j) &\in \mathcal{K}_{N,L}(A_j) \implies \exists (c_j)_{j=1}^b, \\ \forall 1 \leq j \leq b, \quad \Gamma_j(X_j) &= c_j V_j.\end{aligned}\quad (25)$$

The composition gives

$$\begin{aligned}\psi &= \sum_{j=1}^b \Gamma_j(X_j) = \sum_{j=1}^b c_j V_j, \\ \psi &\in \operatorname{span}\{V_j : j = 1, \dots, b\}.\end{aligned}\quad (26)$$

5 Conclusion

This is not a source-error claim. The classification is an open gap: the block-diagonal intersection identity and its periodic-chain iteration have not yet been proved in Lean. Once that theorem is available, issue #905 supplies the structural decomposition needed for issue #870, and the final degenerate parent-Hamiltonian ground-space equality follows by applying the one-block uniqueness theorem to each block.

References

- [CPGSV21] J. Ignacio Cirac, David Pérez-García, Norbert Schuch, and Frank Verstraete. Matrix product states and projected entangled pair states: Concepts, symmetries, theorems. *Reviews of Modern Physics*, 93(4):045003, 2021.
- [PGVWC07] David Pérez-García, Frank Verstraete, Michael M. Wolf, and J. Ignacio Cirac. Matrix product state representations. *Quantum Information and Computation*, 7(5–6):401–430, 2007.